Macroeconomics I — MPE Insper

Practice list: General Equilibrium, Money and Fiscal Policy — qualifier level (Listas 5–6)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

Block 1 — General equilibrium, hours and taxes

(Kurlat ch. 9; Romer ch. 2 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) *General equilibrium × labor supply.* A two-period economy has a fixed capital stock K and no investment. Households have $u(c) + \psi \ln l$ each period, $u(c) = c^{1-\sigma}/(1-\sigma)$, $\sigma > 0$, $\psi > 0$, and one unit of time, $l = 1 - L$; firms produce $Y = AK^{\alpha}L^{1-\alpha}$ competitively, $0 < \alpha < 1$. At the equilibrium hours $L \in (0,1)$, the elasticity $d\ln L/d\ln A$ is:

A. $1/\big[L/(1-L) + \alpha\big]$

B. $(1-\sigma)/\big[L/(1-L) + \alpha + (1-\alpha)\sigma\big]$

C. $1/\big[L/(1-L) + 1\big]$

D. $(1-\sigma)/\big[L/(1-L) + 1\big]$

b) (3 points) *General equilibrium × growth.* In Romer's continuous-time Ramsey model with CRRA coefficient $\sigma > 0$, discount rate ρ, population growth $n \geq 0$, technology growth $g > 0$, depreciation $\delta > 0$, $\rho - n - (1-\sigma)g > 0$ and $f(\tilde{k}) = \tilde{k}^{\alpha}$, $0 < \alpha < 1$, the steady-state gross saving rate is:

A. $\alpha(n+g+\delta)/(\rho+\delta+\sigma g)$

B. $\alpha(n+g)/(\rho+\delta+g)$

C. $\alpha(n+g+\delta)/(\rho+\delta+g/\sigma)$

D. $\alpha(n+g+\delta)/(\rho+\delta+g)$

c) (2 points, T or F) *General equilibrium × fiscal policy.* In the Ramsey model on its balanced growth path, a permanent, unanticipated rise in government purchases per effective worker, financed by lump-sum taxes, lowers consumption one for one on impact and leaves capital unchanged forever.

d) (2 points, T or F) *General equilibrium × taxes.* In the Ramsey model, a permanent tax at rate $\tau_K \in (0,1)$ on capital income, whose revenue is rebated lump-sum, leaves the steady-state capital stock unchanged, because the rebate gives the revenue back.

Question 2. (15 points) Discursive. *General equilibrium × labor supply × taxes.* An infinite-horizon economy without growth: households maximize $\int_0^{\infty} e^{-\rho t}[\ln c + \psi \ln(1-h)]\,dt$, own the capital and rent it to firms; firms produce $Y = K^{\alpha}h^{1-\alpha}$; capital depreciates at rate δ; labor income is taxed at rate τ and the revenue is rebated lump-sum. Parameters: $0 < \alpha < 1$, $\rho > 0$, $\delta > 0$, $\psi > 0$, $0 \leq \tau < 1$.

a) (3 points) Derive the household's optimality conditions (intertemporal and intratemporal), the firm's conditions and the government budget, and write the equilibrium as a dynamic system in (K, c). Show that hours are a well-defined function $h(K, c)$ and find its elasticities.

Solution:

b) (3 points) Find the steady state. Which ratios and prices does the tax leave unchanged? Find steady-state hours in closed form and show that they are unique.

Solution:

c) (3 points) The tax rises permanently and unexpectedly from τ_0 to $\tau_1 \in (\tau_0, 1)$. Find the long-run changes in h, K, Y, C, r and w in closed form, and compare the fall in hours with the fall in the net-of-tax rate $1 - \tau$.

Solution:

d) (3 points) Characterize the transition. Linearize the system in logs at the new steady state, show that it is a saddle, and find the eigenvalues and the half-life in closed form. Find the impact responses of c and h to first order, compare them with the long-run changes, and draw the paths of $\ln K$, $\ln c$ and $\ln h$ (deviations from the old steady state).

Solution:

e) (3 points) *Parameter analysis.* Sign the effects of τ, ψ and δ on h^*; on K^*, Y^*, C^*; on K^*/Y^*, C^*/Y^*, r^*, w^*; and on the speed of convergence. Prove each sign, give the limits as $\tau \to 1$, $\psi \to \infty$ and $\psi \to 0$, and explain why the speed changes.

Solution:

Block 2 — Money, inflation and fiscal policy

(Kurlat chs. 10–11; Romer ch. 11 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Money demand × welfare.* Baumol–Tobin: a household holds its income $Y > 0$ in a bond paying the nominal rate $i > 0$, spends it evenly over the period and withdraws cash in N equal trips at a cost $F > 0$ each (N continuous). The welfare cost of $i > 0$, measured as the Bailey area $\int_0^i m(x)\,dx - i\,m(i)$ under the money demand $m(i)$ this problem generates, is:

A. $\sqrt{FYi/4}$

B. $\sqrt{2FYi}$

C. $\sqrt{FYi/8}$

D. $\sqrt{FYi/2}$

b) (3 points) *Money × expectations.* Cagan with perfect foresight and no bubbles, in logs: $m_t - p_t = -a(p_{t+1} - p_t)$, $a > 0$. Money has been constant; at $t = 0$ the central bank credibly announces a permanent rise of $\theta > 0$ in the log money supply from date $k \geq 1$ on. On impact p_0 rises by:

A. $\left[1 - \left(\frac{a}{1+a}\right)^k\right]\theta$

B. $\frac{1}{1+a}\left(\frac{a}{1+a}\right)^k\theta$

C. $\left(\frac{a}{1+a}\right)^k\theta$

D. $\frac{1}{1+a}\left[1 - \left(\frac{a}{1+a}\right)^k\right]\theta$

c) (2 points, T or F) With money demand as in b) and money growth long at μ_1, a credible announcement at $t = 0$ that money growth will fall permanently to $\mu_2 < \mu_1$ after date $T \geq 1$ leaves inflation at μ_1 until money growth falls.

d) (2 points, T or F) *Money × taxes.* With nominal interest income taxed at $\tau \in (0, 1)$ and a constant after-tax real return, the nominal rate rises by more than one point per point of expected inflation, and the before-tax real rate $i - \pi^e$ rises with π^e.

Question 4. (15 points) Discursive. *Money × fiscal policy × expectations: unpleasant arithmetic.* Real money demand and debt, as shares of GDP, are $m = \kappa e^{-a\pi}$ ($\kappa, a > 0$) and $b_{t+1} = (1+r)b_t + d - S_t$,

with $r > 0$, primary deficit $d \geq 0$ and $b_0 > 0$; output does not grow, so seigniorage at constant inflation $\pi \geq 0$ is $S = \pi m$. The seigniorage S_0 that would hold debt at b_0 is below the largest steady-state seigniorage (part a). As in the Brazilian debates of the 1980s, the central bank tightens and the Treasury does not adjust: for $T \geq 1$ periods ($t = 0, \ldots, T-1$) seigniorage is $S_1 \in (0, S_0)$; from T on, debt is held at b_T.

a) (3 points) Characterize the steady states: the link between debt, deficit and seigniorage; the largest seigniorage; how many inflation rates raise a given S, and which to select and why; and the inflation rate π_0 if debt is stabilized at once, with bounds in terms of S_0 and κ.

Solution:

b) (3 points) Find debt at T and the seigniorage S_2 needed from T on in closed form, and describe debt during the tight phase. Characterize inflation during (π_1) and after (π_2) the tight phase; when does π_2 exist?

Solution:

c) (3 points) Compare π_2 with π_0, and compare the r-discounted average inflation $\bar{\pi} \equiv \frac{r}{1+r}\sum_{t\geq 0}(1+r)^{-t}\pi(t)$ under this policy and under immediate stabilization. Find in closed form the length T^* of the tight phase beyond which no steady state exists.

Solution:

d) (3 points) Now money demand is forward-looking, $m_t - p_t = -a(p_{t+1} - p_t)$ in logs, with perfect foresight and no bubbles. Money has long grown at μ_1; at $t = 0$ it is credibly announced that $m_t - m_{t-1} = \mu_1$ for $t \leq T$ and $\mu_2 > \mu_1$ for $t > T$. Find the price level, its jump on announcement and inflation at every date; characterize inflation before T (bounds, monotonicity, half-life) and the timing of the fall in real balances; read it with $\mu_1 = \pi_1$, $\mu_2 = \pi_2$.

Solution:

e) (3 points) *Parameter analysis.* Sign, with proofs, the effects of S_1, T, r, d (or b_0), κ, a on S_2, π_2, $\pi_2 - \pi_1$, T^* in a table; give the limits $T \to 0$, $S_1 \to S_0$, $T \to T^*$, $r \to 0$, $a \to 0$ with their reading; draw π_2 against T for two values of S_1.

Solution:

Block 3 — Integrative: growth, interest rates, money and seigniorage

(Kurlat chs. 9–11; Romer ch. 11 • 25 points)

Question 5. (10 points) Objective items.

Items a)–d) use the economy of Question 6 and compare balanced paths, on which i, r and π are constant. A closed economy receives the endowment $Y_t = Y_0(1+g)^t$, $g \geq 0$. The representative household maximizes $\sum_{t\geq 0} \beta^t c_t^{1-\sigma}/(1-\sigma)$ ($\ln c_t$ if $\sigma = 1$), with $\sigma > 0$, $0 < \beta < 1$ and $\beta(1+g)^{1-\sigma} < 1$, and trades one-period real and nominal bonds. Real money demand is $M_t/(P_t Y_t) = \kappa e^{-a i_t}$, $\kappa, a > 0$; the Fisher equation holds exactly, $1 + i = (1+r)(1+\pi)$; policy keeps $i \geq 0$.

a) (3 points) *General equilibrium × money.* Let $\sigma = 1$. The central bank adjusts the money supply so that inflation equals its target $\pi^* > 0$. The equilibrium nominal interest rate is:

A. $i = (1+g)(1+\pi^*)/\beta - 1$

B. $i = \beta(1+\pi^*) - 1$

C. $i = (1+\pi^*)/\beta - 1$

D. $i = \beta(1+g)(1+\pi^*) - 1$

b) (3 points) *Seigniorage × interest rates.* A government finances itself by printing money, as Brazil's did before the 1994 Real Plan. Let $g = 0$ and write $r > 0$ for the equilibrium real rate. The money stock grows at a constant rate $\mu > 0$ and seigniorage is $S_t = (M_{t+1} - M_t)/P_t$, so $S/Y = \mu\kappa e^{-ai}$ on a balanced path. The money growth rate that maximizes S/Y is:

A. $\mu^* = 1/a$

B. $\mu^* = 1/[a(1+r)]$

C. $\mu^* = 1/a - r$

D. $\mu^* = 1/[a(1+r)] - r$

c) (2 points, T or F) *General equilibrium × money × welfare.* In this endowment economy a permanent rise in money growth leaves the real interest rate unchanged and therefore has no real effects.

d) (2 points, T or F) *Growth × money.* Under a constant money-growth rule, faster trend growth always lowers the nominal interest rate because it lowers inflation.

Question 6. (15 points) Discursive. *Integrative: growth × equilibrium interest rates × money demand × seigniorage × welfare.* An endowment economy produces $Y_t = Y_0(1+g)^t$, $Y_0 > 0$, $g \geq 0$. A representative household maximizes $\sum_{t\geq 0} \beta^t c_t^{1-\sigma}/(1-\sigma)$ (logarithmic at $\sigma = 1$), with $\sigma > 0$ and $\beta \in (0,1)$, and trades one-period real and nominal bonds, both in zero net supply. Real money demand is the reduced form $M_t/(P_tY_t) = \kappa e^{-ai_t}$, $\kappa, a > 0$, where i_t is the nominal interest rate, and the Fisher relation holds exactly: $1 + i_t = (1+r_t)(1+\pi_{t+1})$. The government hands the new money to the household as a lump-sum transfer; with money growing at rate μ and new money valued at today's prices, seigniorage is $S_t/Y_t = \mu\, M_t/(P_tY_t)$. Assume $\beta(1+g)^{1-\sigma} < 1$ and study steady states (constant r, i, π, μ) with $i \geq 0$. Brazil adopted inflation targeting in 1999 and has lived with high real interest rates; the items ask what this economy says about both. All answers are symbolic.

a) (3 points) Derive the Euler equation and find the equilibrium real interest rate. Show that $\beta(1+g)^{1-\sigma} < 1$ is the condition for household wealth to be finite and say what it implies for r and g. Under a money-growth rule $M_{t+1} = (1+\mu)M_t$, find steady-state inflation, the nominal rate and the admissible range of μ.

Solution:

b) (3 points) Compare the money-growth rule with inflation targeting, in which the central bank sets money growth so that $\pi = \pi^*$, with $\pi^* > 0$. For each regime find i and μ, and prove for which σ faster trend growth raises the nominal rate. Decompose the response into the forces at work.

Solution:

c) (3 points) Under the money-growth rule, find the money growth rate μ^* that maximizes seigniorage, check the second-order condition, prove that μ^* is the unique admissible maximizer, find the maximum revenue $s_{\max}$ in closed form, and determine how trend growth moves the peak.

Solution:

d) (3 points) Measure the welfare cost of a steady state with nominal rate i, relative to the Friedman rule $i = 0$, by the Bailey area $W(i)$: the area under the inverse money-demand curve between the balances held at i and at $i = 0$. Find $w(i) \equiv W(i)/Y$ in closed form, prove that it rises with i and characterize it for small and large i. Find the inflation and money growth rates that implement the Friedman rule. Under the target π^*, with nominal rate i_T, find the cost per unit of seigniorage, average and marginal. Draw the demand curve and the cost at the target.

Solution:

e) (3 points) *Parameter analysis.* Sign the effects of σ, g, β, a and κ on r, on i under each regime, on μ^* and $s_{\max}$, and on the cost at the target, $w_T \equiv w(i_T)$, proving each sign or giving the condition that decides it. Study $\sigma \to 0$, $\sigma = 1$, $g \to 0$ and $a \to 0$, and read the results for Brazil.

Solution: